

Handwritten exercises on GLMs - Handout

1 Exercise 1

Show that the Binomial distribution and the Gamma distribution belong to the exponential family of distributions, by identifying θ , A , ϕ , and the functions $b()$ and $c()$ (cf. p. 53 of the course slides). Using the properties of the distributions of the exponential family, give the expectation and the variance in each case.

Notation:

- If Y follows a binomial distribution $\mathcal{B}(m, p)$, then

$$P(Y = y; m, p) = \binom{m}{y} p^y (1-p)^{m-y}, \quad \text{for } y = 0, \dots, m.$$

From the pmf above, we can write

$$\begin{aligned} \mathbb{P}(Y = y; m, p) &= \binom{m}{y} \left(\frac{p}{1-p} \right)^y (1-p)^m \\ &= \exp \left\{ \ln \binom{m}{y} + y \ln \left(\frac{p}{1-p} \right) + m \ln(1-p) \right\} \\ &= \exp \left\{ y \ln \left(\frac{p}{1-p} \right) + m \ln(1-p) + \ln \binom{m}{y} \right\} \end{aligned}$$

From this, we deduce $\theta = \ln \frac{p}{1-p}$. Also, $b(\theta) = -m \ln(1-p)$.

From $\theta = \ln \frac{p}{1-p}$ we obtain

$$\begin{aligned} \exp \theta &= \frac{p}{1-p} \\ \exp \theta - p \exp \theta &= p \\ \exp \theta &= p(1 + \exp \theta) \\ p &= \frac{\exp \theta}{1 + \exp \theta} \end{aligned}$$

and therefore $b(\theta) = -m \ln\left(\frac{1}{1+\exp \theta}\right) = m \ln(1 + \exp \theta)$.

At this point, we have :

$$\mathbb{P}(Y = y; m, p) = \exp\{y\theta - b(\theta) + \ln \binom{m}{y}\},$$

which leaves $\phi = 1$, $A = 1$ and $c(y, \frac{\phi}{A}) = \ln \binom{m}{y}$

$$\begin{aligned} - \theta &= \ln \frac{p}{1-p} \\ - b(\theta) &= m \ln(1 + \exp \theta) \\ - \phi &= 1 \\ - A &= 1 \\ - c\left(y, \frac{\phi}{A}\right) &= \log \binom{m}{y} \end{aligned}$$

This gives us $\mathbb{E}(Y|\theta) = b'(\theta) = m \frac{\exp \theta}{1+\exp \theta} = mp$ and $\text{Var}(Y|\theta) = m \frac{b''(\theta)\phi}{A} = \frac{\exp \theta}{(1+\exp \theta)^2} = mp(1-p)$

- If Y follows a Gamma distribution $\Gamma(\mu, \nu)$, then

$$f(y; \mu, \nu) = \frac{\nu \exp(-\nu y/\mu) (\nu y/\mu)^{\nu-1}}{\mu \Gamma(\nu)}, \quad \text{for } y > 0.$$

From the pdf above, we can write

$$\begin{aligned} f(y; \mu, \nu) &= \exp \left(\ln \frac{\nu \exp(-\nu y/\mu) (\nu y/\mu)^{\nu-1}}{\mu \Gamma(\nu)} \right) \\ &= \exp \left(\ln \nu - \nu \frac{y}{\mu} + \nu \ln \nu + \nu \ln \frac{y}{\mu} - \ln \nu - \ln \frac{y}{\mu} - \ln \mu - \ln \Gamma(\nu) \right) \\ &= \exp \left(-\nu \left(y \frac{1}{\mu} - \ln \frac{1}{\mu} \right) + \nu \ln \nu + (\nu - 1) \ln y - \ln \Gamma(\nu) \right) \end{aligned}$$

- $\theta = \frac{1}{\mu}$
- $b(\theta) = \ln\left(\frac{1}{\mu}\right) = \ln \theta$

- $\phi = \frac{1}{\nu}$
- $A = -1$
- $c\left(y, \frac{\phi}{A}\right) = -\frac{A}{\phi} \ln\left(-\frac{A}{\phi}\right) + \left(-\frac{A}{\phi} - 1\right) \ln y - \ln \Gamma\left(-\frac{A}{\phi}\right)$

This gives us $\mathbb{E}(Y|\theta) = b'(\theta) = \frac{1}{\theta} = \mu$ and $\text{Var}(Y|\theta) = \frac{b''(\theta)\phi}{A} = \frac{1}{\theta^2\nu} = \frac{\mu^2}{\nu}$

2 Exercise 2

The beta distribution is used for situations where the measured variable takes values between 0 and 1, for example proportions.

We let Y follow a beta distribution with parameters μ and ϕ , where μ is the parameter of interest and ϕ is a nuisance parameter. The density function of Y is

$$f(y) = \frac{\Gamma(\phi)}{\Gamma(\phi\mu) + \Gamma(\phi(1-\mu))} y^{\phi\mu-1} (1-y)^{\phi(1-\mu)-1}, \quad \text{for } 0 < y < 1.$$

Does this distribution belong to the exponential family? Justify your answer, and if yes deduce $E(Y)$ and $\text{Var}(Y)$.

We write

$$\begin{aligned} f(y) &= \exp \{ \ln \Gamma(\phi) - \ln \Gamma(\phi\mu) - \ln \Gamma(\phi(1-\mu)) + (\phi\mu - 1) \ln y + (\phi(1-\mu)) \ln(1-y) \} \\ &= \exp \{ (\phi\mu - 1) \ln y + (\phi(1-\mu)) \ln(1-y) + \ln \Gamma(\phi) - \ln \Gamma(\phi\mu) - \ln \Gamma(\phi(1-\mu)) \}. \end{aligned}$$

We therefore cannot express $f(y)$ as

$$f(y) = \exp \left\{ A \frac{y\theta - b(\theta)}{\phi} + c\left(y, \frac{\phi}{A}\right) \right\}.$$

In particular, the term $y\theta$ cannot be identified.

With this parametrisation, the beta distribution does not belong to the exponential family.

3 Exercise 3

- Let $f(y; \theta)$ be the probability density function of a random variable Y , for a given parameter θ . Without using the exponential family structure, show that for $U(\theta; Y) = \frac{\partial}{\partial \theta} \log f(Y; \theta)$

a) $\mathbb{E}[U(\theta; Y)] = 0,$

$$\begin{aligned}\mathbb{E}[U(\theta; Y)] &= \mathbb{E}\left[\frac{\partial}{\partial \theta} \log f(Y; \theta)\right] \\ &= \mathbb{E}\left[\frac{1}{f(Y; \theta)} \frac{\partial}{\partial \theta} f(Y; \theta)\right] \\ &= \int \frac{1}{f(y; \theta)} \left[\frac{\partial}{\partial \theta} f(y; \theta)\right] f(y; \theta) dy \\ &= \int \frac{\partial}{\partial \theta} f(y; \theta) dy \\ &= \frac{\partial}{\partial \theta} \int f(y; \theta) dy \quad \text{if the support is compact} \\ &= \frac{\partial}{\partial \theta} (1) \quad \text{given that } f(y; \theta) \text{ is a density} \\ \mathbb{E}[U(\theta; Y)] &= 0\end{aligned}$$

b) $\mathbb{E} \left[\frac{\partial U(\theta; Y)}{\partial \theta} \right] = -\mathbb{E} [U^2(\theta; Y)].$

$$\begin{aligned} \mathbb{E} \left[\frac{\partial U(\theta; Y)}{\partial \theta} \right] &= \mathbb{E} \left[\frac{\partial^2}{\partial \theta^2} \log f(Y; \theta) \right] = \mathbb{E} \left\{ \frac{\partial}{\partial \theta} \left[\frac{1}{f(Y; \theta)} \frac{\partial}{\partial \theta} f(Y; \theta) \right] \right\} \\ &= \mathbb{E} \left\{ \frac{1}{f(Y; \theta)^2} \left[f(Y; \theta) \frac{\partial^2}{\partial \theta^2} f(Y; \theta) - \left(\frac{\partial}{\partial \theta} f(Y; \theta) \right)^2 \right] \right\} \\ &= \mathbb{E} \left[\frac{1}{f(Y; \theta)} \frac{\partial^2}{\partial \theta^2} f(Y; \theta) \right] - \mathbb{E} \left[\frac{1}{f(Y; \theta)} \frac{\partial}{\partial \theta} f(Y; \theta) \right]^2 \\ &= \int \frac{\partial^2}{\partial \theta^2} f(y; \theta) \, dy - \mathbb{E} \left[\frac{\partial}{\partial \theta} \log f(Y; \theta) \right]^2 \\ &= \frac{\partial^2}{\partial \theta^2} \int f(y; \theta) \, dy - \mathbb{E} [U^2(\theta; Y)] \\ \mathbb{E} \left[\frac{\partial U(\theta; Y)}{\partial \theta} \right] &= -\mathbb{E} [U^2(\theta; Y)] \end{aligned}$$

- What condition do we have to impose on the support of Y to prove these two statements?

The support of the random variable Y needs to be compact, so that in particular the integral and the partial derivative operators can be exchanged.

4 Exercise 4

In this exercise, we consider the problem of Maximum Likelihood estimation of the parameters of a Poisson random variable with a log link, i.e :

$$g(\mu_i) = \ln \{ \mathbb{E} [Y_i \mid \mathbf{X}_i = \mathbf{x}_i] \} = \mathbf{x}_i^T \boldsymbol{\beta}$$

- a) Write the estimating (score) equations for the proposed model.

Start by writing :

$$\mathcal{L}(\boldsymbol{\beta}; \mathbf{y}, \mathbf{X}) = \prod_{i=1}^n \frac{\exp(y_i x_i^T \boldsymbol{\beta}) \exp(-\exp(x_i^T \boldsymbol{\beta}))}{y_i!}$$

$$\ell(\boldsymbol{\beta}; \mathbf{y}, \mathbf{X}) = \sum_{i=1}^n [y_i x_i^T \boldsymbol{\beta} - \exp(x_i^T \boldsymbol{\beta}) - \ln(y_i!)]$$

The score function is:

$$\begin{aligned} \frac{\partial}{\partial \boldsymbol{\beta}} \ell(\boldsymbol{\beta}; \mathbf{y}, \mathbf{X}) &= \sum_{i=1}^n [y_i x_i - \exp(x_i^T \boldsymbol{\beta}) x_i] \\ &= \sum_{i=1}^n [y_i - \exp(x_i^T \boldsymbol{\beta})] x_i \end{aligned}$$

b) Show that

$$\mathcal{J}(\boldsymbol{\beta}; \mathbf{y}) = \sum_{i=1}^n \exp\{\mathbf{x}_i^T \boldsymbol{\beta}\} \mathbf{x}_i \mathbf{x}_i^T.$$

From $\mathcal{J}(\boldsymbol{\beta}; \mathbf{y}) = -\mathbb{E} \left(\frac{\partial^2}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}^T} \ell(\boldsymbol{\beta}; \mathbf{y}, \mathbf{X}) \right)$ and what we found in a),

$$\begin{aligned} \frac{\partial^2}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}^T} \ell(\boldsymbol{\beta}; \mathbf{y}, \mathbf{X}) &= \frac{\partial}{\partial \boldsymbol{\beta}} \sum_{i=1}^n [y_i - \exp(x_i^T \boldsymbol{\beta})] x_i \\ &= - \sum_{i=1}^n [\exp(x_i^T \boldsymbol{\beta})] x_i x_i^T \\ \Leftrightarrow \mathcal{J}(\boldsymbol{\beta}; \mathbf{y}) &= \sum_{i=1}^n [\exp(x_i^T \boldsymbol{\beta})] x_i x_i^T \end{aligned}$$

c) Write the formula for the Newton-Raphson algorithm.

$$\boldsymbol{\beta}^{(k)} = \boldsymbol{\beta}^{(k-1)} + \left\{ \sum_{i=1}^n [\exp(x_i^T \boldsymbol{\beta})] x_i x_i^T \right\}^{-1} \left\{ \sum_{i=1}^n [y_i - \exp(x_i^T \boldsymbol{\beta})] x_i \right\}$$